

기량 4. / 한쪽 분포 이론 2.

* χ^2 (카이제곱 분포)

* t (t 분포)

* C.L.T

* 카이제곱 분포

• $Z \sim N(0, 1)$ 이면,

$Z^2 \sim \chi^2(\underline{1})$ 이다

• $X_1, X_2, \dots, X_p \sim \underline{\text{iid}} N(0, 1)$

(identically,

independently, distributed)

이면, $X_1^2 + \dots + X_p^2 \sim \chi^2(p)$

Graph



* t-분포

- Z : r. v. (random variable)
 $\sim N(0, 1)$
- V 는 자유도 ν 인 χ^2 분포를 따른다.

$$Z \sim N(0, 1)$$

$$V \sim \chi^2(\nu)$$

$$X = \frac{Z}{\sqrt{V/\nu}} \sim t\text{-분포}(\nu)$$

<특징>

$$E(X) = 0$$

$$\text{Var}(X) = \frac{\nu}{\nu-2} \quad (\nu > 2)$$

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$> 1$ , 이라는 것은

Standard. N과 비교

$\Rightarrow$  더 큰 분산성향  $\uparrow$



if  $\nu \rightarrow \infty$ ,

t-분포  $\xrightarrow{d}$   $N(0,1)$

## \* F-분포

let  $A, B$  independent  
r. v

$$A \sim \chi^2(r_1)$$

$$B \sim \chi^2(r_2)$$

$$F = \frac{A/r_1}{B/r_2} \sim F(r_1, r_2)$$

# \* 중심극한정리 (Central Limit Theorem)

< 분포수렴 >

$X_1, X_2, \dots, X_n, X_{n+1}, \dots$  이 독립 변수

$X$ 의 분포로 수렴한다

(Convergence in distribution)

$$X_n \xrightarrow{d} X \text{ 라고 쓰고}$$

$$\lim_{n \rightarrow \infty} P(X_n \leq x) = P(X \leq x)$$

for  $F_X$ 가 연속인 모든 점  $x$ .

즉, 누적분포함수가 수렴할 때,

분포가 수렴한다고 한다.

(C.L.T) 독립변수  $X_1, X_2, X_3, \dots, X_n, \dots$

의 평균이  $\mu$ 이고, 분산이  $\sigma^2$ 인  $i.i.d$

독립변수라 할 때,

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \text{에 대해}$$

$$\frac{\sqrt{n} (\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0, 1)$$

$$Y_i = \mu + e_i,$$

$$e_i \sim \text{i.i.d. } N(0, \sigma^2)$$

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표본평균

$$E(\bar{y}) = \mu$$

" 분산

$$\text{Var}(\bar{y}) = \frac{\sigma^2}{n}$$

→  $\bar{y}$ 의 분포?

<  $\sigma^2$  : known

$\sigma^2$  : unknown

$$\Rightarrow \tilde{\bar{y}} \sim \boxed{N}(\mu, \frac{\sigma^2}{n})$$

(case1)  $\sigma^2$ : known,  $e_i \sim i.i.d(N(0, \sigma^2))$

Let,  $\underbrace{Z_i}_{\text{S.N}} = \left( \frac{e_i}{\sigma} \right) \sim N(0, 1)$

$$\underline{Z_1^2} + \underline{Z_2^2} + \dots + \underline{Z_n^2} \sim \chi^2(n)$$

$$= \left( \frac{e_1}{\sigma_1} \right)^2 + \dots + \left( \frac{e_n}{\sigma_n} \right)^2 \sim \chi^2(n)$$

$$= \sum_{i=1}^n Z_i^2 \sim \chi^2(n)$$

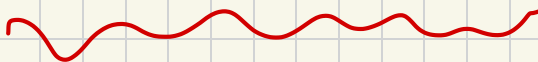
$$\boxed{\frac{\sum e_i^2}{\sigma^2} \sim \chi^2(n)}$$

$$e_i = y_i - \mu,$$

sample equation

$$\Rightarrow \hat{e}_i = y_i - \hat{\mu}$$

$$\Rightarrow \frac{\sum \hat{e}_i^2}{\sigma^2} \sim \chi^2(n-1)$$



Q)  $\sigma^2$  이 unknown?

표본 분산과 관련한 문제.

$$\hat{\sigma}^2$$

$$\Rightarrow \hat{e}_i = y_i - \hat{\mu}$$

$$\frac{\pm \hat{e}_i^2}{\sigma^2} \sim \chi^2(n-1) \quad \text{--- (A)}$$

과 관련한 H?

$$\hat{\sigma}^2 = \frac{\pm \hat{e}_i^2}{n-1}$$

$$\frac{(n-1) \hat{\sigma}^2}{\sigma^2} \sim \chi^2(n-1)$$

$$\star \quad E(\hat{\sigma}^2) = \sigma^2 \quad \text{일지?}$$

$\Rightarrow \hat{\sigma}^2$ 은  $\sigma^2$ 의 unbiased Estimator?

(\*) Note

일어  $X$ ,

$X \sim \chi^2(p)$  라 하자.

$$E(X) = p.$$

$$\text{var}(X) = 2p$$

Based on  $\chi^2$  정리,

$$E\left(\frac{\cancel{(n-1)} \hat{\sigma}^2}{\cancel{\sigma^2}}\right) = \cancel{n-1}$$

$$\Rightarrow E(\hat{\sigma}^2) = \sigma^2 \quad \checkmark$$

$$\text{var}\left(\frac{\cancel{(n-1)} \hat{\sigma}^2}{\cancel{\sigma^2}}\right) = 2 \cdot (n-1)$$

$$\frac{(n-1)^2}{\sigma^4} \text{var}(\hat{\sigma}^2) = 2(n-1)$$

$$\text{var}(\hat{\sigma}^2) = \frac{2\sigma^4}{n-1} \quad \checkmark$$

< Summary >

$$\frac{(n-1) \hat{\sigma}^2}{\sigma^2} \sim \chi^2 (n-1)$$

$$\sqrt{\hat{\sigma}^2} = \frac{\pm \hat{e}_i^2}{\underbrace{(n-1)}_{\text{"}}}$$

$$E(\hat{\sigma}^2) = \sigma^2$$

$$\text{Var}(\hat{\sigma}^2) = \frac{2\sigma^4}{n-1}$$

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